

BAND GAP ENERGY CHARACTERISTICS IN MATERIALS:

Band gap energy is one of the most fundamental parameters in determining the electronic and optoelectronic properties of a material. The band gap value is a measure of the energy jump that electrons must make in order to actively conduct electricity. The greater the band gap value, the greater the energy required by electrons to conduct electricity.

The difference in band gap values in various materials affects the ability of the material to conduct electricity or interact with light. Therefore, understanding the characteristics of band gap energy is very important in the development of various devices such as semiconductors, photovoltaics, optical sensors, and light-emitting diodes (LEDs).

Band gap is the energy difference between the highest occupied electron energy level (in the valence band) and the lowest empty electron energy level (in the conduction band). The magnitude of the band gap value is the main determinant of whether a material behaves as a conductor, semiconductor, or insulator.

Material Classification Based on Band Gap Energy

In general, materials are classified based on the size of their band gap energy as follows:

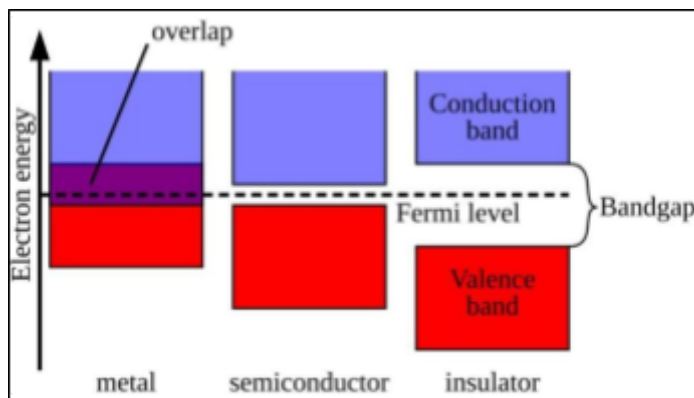


Figure 1.1 Semiconductor band gap

Conductor

It does not have a significant band gap; the valence and conduction energy bands meet. Electrons can move freely without any additional external energy. Therefore, conductors such as copper (Cu) and aluminum (Al) are very efficient in conducting electricity.

Insulator

Having a large band gap (>5 eV). Electrons are tightly bound in the valence band and require large energy to move to the conduction band, making them almost incapable of conducting electricity. Examples of insulators are glass, ceramics, and mica.

Semiconductor

Having a medium band gap (around $0.5 - 3$ eV). At room temperature, some electrons are

able to move to the conduction band and provide medium conductivity properties. Eg; Silicon (Si): 1.1 eV (indirect band gap), Gallium Arsenide (GaAs): 1.42 eV (direct band gap), Gallium Nitride (GaN): 3.4 eV (direct band gap)

Electrical conductivity is the intrinsic ability of a material to conduct electric current when given a potential difference. Materials are grouped into conductors, semiconductors, and insulators based on their conductivity levels. Conductors offer low resistance to current flow, while insulators offer very high resistance. Semiconductor materials themselves act as insulators at very low temperatures and act as conductors at room temperature.

To explain the conductivity of a material, the concept of energy bands is often used. There are two energy bands, namely the valence band and the conduction band. The valence band is an energy band that may be filled by electrons from a solid until complete/full, while the conduction band is an energy band that is another place that will be filled by electrons after the valence band is full. At a temperature of 0 K, the conduction band is partially filled for conductor materials, while insulators and semiconductors have no electrons filling the conduction band.

Band gap energy is one of the distinguishing indicators between semiconductor materials and other materials. Band gap energy is one of the typical properties of semiconductor materials that describes how much minimum energy is needed for electrons to move from the valence energy level to the conduction energy level. Information about the size of this energy gap is very useful in determining the appropriate approach or treatment of the material in various technological applications.

BAND GAP ENERGY CHARACTERISTICS IN MATERIALS

To determine the energy band gap of a thin layer of amorphous silicon that has been deposited, the common method used is the Tauc Plot method. (Tauc 1972)

$$(ah\nu) = A (h\nu - E_g)^n \quad (1.1)$$

$$\alpha = - \ln(T) / d \quad \dots \quad (1.2)$$

In this context, the absorption coefficient is calculated for a thin film using the transmittance data obtained from UV-Vis measurements. The variable 'd' represents the thickness of the deposited film, 'hν' is the photon energy, 'A' is a constant, 'n' is 2 for indirect band gap materials, and 'E_g' is the band gap energy. The band gap energy of a thin film can be determined by finding the linear portion of the Tauc Plot curve, extrapolating it until it intersects the X-axis; the intersection point represents the band gap energy value.

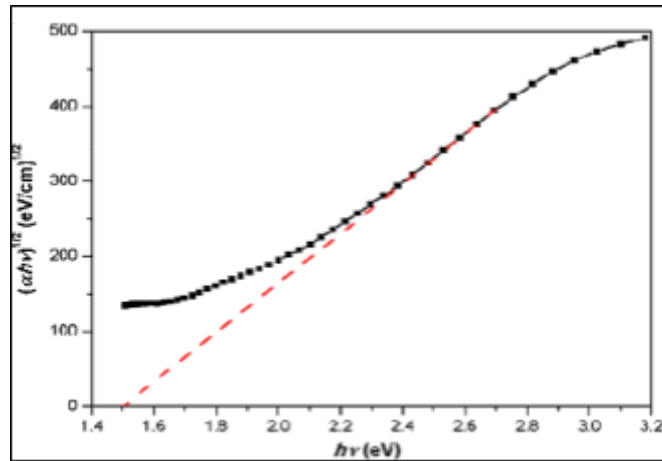


Figure 1.2: This schematic diagram, as described by Yuan et al. (2014)

Figure 1.2 is an example of one of the measurements of the energy gap of amorphous silicon thin films using the Tauc Plot method. Through a linear line drawn to intersect the x-axis, it can be determined that the energy band gap of the film is around 1.5 eV. A decrease in the energy band gap will be accompanied by an increase in the absorption level of the film. Thin films can only absorb photons with higher energy than their energy gap, while photons with lower energy are reflected. Therefore, the energy band gap of the film is an important parameter in the absorption and transmittance aspects that can affect the efficiency of solar cells from the sunlight energy entering the film.

Amorphous silicon is a semiconductor material that is widely studied and used in various thin-film technologies. One of its main applications is in solar panels (photovoltaic or PV). However, the efficiency of solar panels using amorphous silicon is still relatively low. Currently, PV modules based on amorphous silicon with a single-pin structure can only achieve efficiencies of around 6-8%. This One way to make solar cells better is to improve the performance of amorphous silicon (a-Si) solar panels. Amorphous silicon has an energy band gap of around 1.8 eV, which is ideal for absorbing energy from sunlight. The graph shows that the highest efficiency can be achieved with an energy band gap around this value. Theoretically, ideal amorphous silicon solar cells can achieve efficiencies approaching 30%, making it a very promising material for solar cell manufacturing.

External Factors Affecting Band Gap

In addition to intrinsic properties, the band gap can be influenced by:

1. Temperature: The band gap tends to narrow at high temperatures due to expansion of the crystal lattice and increased atomic vibrations.
2. Doping: The addition of foreign atoms can modify the energy levels in the valence and conduction bands (example: n and p doping in semiconductors).

3. Pressure and stress: Can cause changes in crystal structure and affect the bandgap energy.
4. Particle size: In the nanoscale (quantum dot), the band gap can be increased due to the energy quantization effect (quantum confinement).

Band Structure: Direct vs Indirect Band Gap

The fundamental difference between direct and indirect band gap semiconductors lies in the energy band structure and electron transition mechanism.

Direct Band Gap

In direct band gap semiconductors, the minimum position of the conduction band and the maximum of the valence band are in the same momentum (k-space). This means that electrons can move directly from the conduction band to the valence band by emitting photons, without the help of momentum changes. This makes it very efficient in the process of light emission.

Examples: GaAs, GaN, InP.

Indirect Band Gap

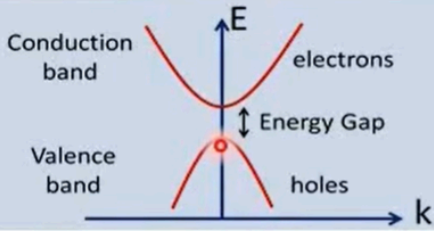
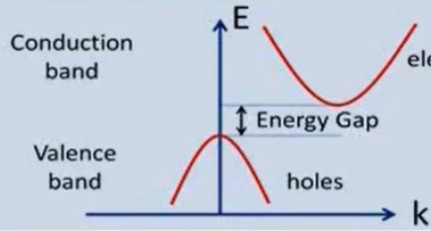
In indirect band gap semiconductors, the conduction band minimum and valence band maximum are not at the same momentum. Electron transitions require additional interactions with phonons (lattice vibrations) to change momentum. This process is inefficient in emitting light, making it unsuitable for optoelectronic applications. Example: Si, Ge.

The energy of electron is given by ,

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

Where, P is momentum , m is mass of electron, h is plank constant and k is propagation constant.

The relation E is directly proportional to k² is equation of parabola.

Direct bandgap Semiconductors	Indirect bandgap Semiconductors
	
Maximum of valence band and minimum of conduction band occur at same momentum values	Maximum of valence band and minimum of conduction band occur at two different momentum values.
Electron making a transition from valence band to conduction band need not undergo any change in its momentum.	In order to make a transition from maximum point in valence band to minimum point in conduction band, the electron requires energy for the change in momentum in addition to the energy gap E_g
The compound semiconductors such as GaAs, are direct gap semiconductors	All elemental semiconductors such as Si, Ge, are indirect gap semiconductors
These direct gap semiconductors are used in LED and Semiconductor Lasers.	Not useful for LEDs and Semiconductor Lasers

• Band Gap Contribution to the Technological Revolution

The development of modern technology is highly dependent on band gap engineering, for example:

- A. Materials with direct band gap are ideal for LED (Light Emitting Diode), photonic sensors and devices. Here the light emission is recombination of electrons and holes (radiative recombination) by direct band gap material due to strong absorption. eg., GaN and InGaN based on materials, energy-efficient LEDs and full-color displays
- B. Materials with indirect gap such as silicon that have an indirect band gap are not used in making LEDs because the light produced is very weak or almost nonexistent. This is why even though silicon dominates the microelectronics industry, it is not used in light-emitting applications.
- C. A new generation of solar cells that combine direct band gap materials and multi-junction structures to capture a wider spectrum of light.

• Importance of refractive index, optical absorption and band gap in semiconducting materials:

- ☐ The refractive index of semiconducting materials is a crucial factor in determining their optical and electronic properties. It is a *dimensionless quantity that describes how much light slows down or bends when it enters the material*. The refractive index is closely related to the energy gap of the material, which determines the threshold for absorption of photons. A correlation between the refractive index and energy gap is significant for the

design and performance of devices such as photonic crystals, wave guides, solar cells, and detectors. The relationship is based on the photo effect studies and the effective dielectric constant of the material.

1. Energy Gap (E_g) and Refractive Index (n) relation

- Refractive index is the ratio of velocity of light in vacuum to that in the material.
- In semiconductors, the refractive index decreases with increase in energy gap.

That can be understood by following Empirical Relations:

(a) Moss Relation

$$n^4 \times E_g = \text{constant}$$

(b) Ravindra Relation

$$n = 4.084 - 0.62 \times E_g$$

(c) Herve–Vandamme Relation

$$n^2 = 1 + [13.6 / (E_g + 3.4)]^2$$

From the above equations, we can say that the *refractive index is inversely proportional to the band gap*.

Narrow band gap $\rightarrow$ High refractive index

Wide band gap $\rightarrow$ Low refractive index

☐ Optical Absorption and in Semiconductors

- Optical absorption occurs when photon energy ($h\nu$) $\geq$ energy gap (E_g).
- Electrons are excited from valence band to conduction band.

Absorption Coefficient:

$$I = I_0 \times e^{(-\alpha x)}$$

Where:

I_0 = incident light intensity

I = transmitted intensity

α = absorption coefficient

x = thickness of semiconductor

2. Energy Gap and Optical Absorption (Tauc Relations)

(a) Direct Band Gap Semiconductors

(Example: GaAs, InP)

$$\alpha \times h\nu = A \times (h\nu - E_g)^{(1/2)}$$

- Strong absorption

- Sharp absorption edge

(b) Indirect Band Gap Semiconductors

(Example: Si, Ge)

$$\alpha \times hv = A \times (hv - E_g)^2$$

- Weak absorption
- Phonon participation required

4. Relation Between Refractive Index and Absorption

Complex Refractive Index:

$$\tilde{n} = n + i k$$

Where:

n = refractive index

k = extinction coefficient

Absorption Coefficient:

$$\alpha = (4 \times \pi \times k) / \lambda$$

Note:

k tells how strongly electrons absorb light at a given wavelength.

α tells how fast light intensity decays as it travels through the semiconductor.

Optical absorbance is directly proportional to the extinction coefficient k , which is the imaginary part of the refractive index.

5. Importance of Refractive Index in Semiconducting Materials

- Controls speed and propagation of light in semiconductors.
- Important for LEDs, laser diodes, photodetectors, and solar cells.
- Helps reduce reflection losses using anti-reflection coatings.
- Inversely related to energy gap, useful for estimating band gap (E_g).
- Used in optical waveguides and photonic devices.
- Useful for optical characterization of materials.

Refractive index is important because it controls light reflection, transmission, confinement, and absorption in optoelectronic devices.

Note:

- Increase in $E_g \rightarrow$ Decrease in $k \rightarrow$ Decrease in Absorbance $\rightarrow$ Decrease in n
- Direct band gap $\rightarrow$ Strong absorption
- Indirect band gap $\rightarrow$ Weak absorption
- Absorption edge gives value of energy gap

Basic material properties needed for semiconductor device fabrication:

Semiconductor materials are an essential part of almost all current electronic devices, from microprocessors and sensors to light-emitting diodes and solar cells. Their unique ability to conduct electric current under certain conditions makes them ideal for switching, amplification and energy conversion functions.

Key Properties of Semiconductors

1. Band Gap Energy ((E_g))

Smaller E_g gives easier conduction, while larger E_g gives better high-temperature and high-voltage performance.

Small band gap → easier conduction (e.g., Germanium)

Large band gap → higher temperature and voltage tolerance (e.g., SiC, GaN) (GaN: ~3.4 eV, wide band gap, useful for high-power devices)

2. Intrinsic Conductivity:

The natural ability of a pure semiconductor to conduct electricity without impurities. Conductivity is very low in pure materials, but can be controlled by adding dopants.

Intrinsic conductivity is important in semiconductor devices because it determines the natural carrier concentration, temperature behavior, leakage current, breakdown voltage, and overall device stability before doping.

① Low intrinsic conductivity at room temperature is required

- So that conductivity can be controlled effectively by doping.
- Ensures proper formation of PN junctions.

② Should increase with temperature in a controlled way

- Helps in temperature sensing devices.
- Excessive intrinsic conduction at high temperature causes leakage current.

③ Low intrinsic carrier concentration improves device performance

- Reduces unwanted current in diodes and transistors.
- Increases breakdown voltage.

④ Wide band gap materials are preferred for power devices

- Very low intrinsic conductivity
 - Can operate at high temperature and high voltage
- (Example: SiC, GaN)

5 Provides basic understanding before doping

→ Device design starts from intrinsic properties.

3. Mobility:

Mobility (μ) is the measure of how fast a charge carrier moves per unit electric field.

$$v_d = \mu E$$

Where:

- v_d = drift velocity
- μ = mobility
- E = electric field

The speed at which charge carriers (electrons and holes) move through the crystal lattice under an electric field.

Higher mobility → faster switching speeds in transistors. Units: $\text{cm}^2/(\text{V}\cdot\text{s})$

Example: Electrons in GaAs have higher mobility than in silicon, making GaAs useful for high-frequency applications.

4. Thermal Conductivity

Thermal conductivity in semiconductors is crucial for managing heat in electronic devices, influencing their performance and reliability.

Thermal conductivity in semiconductors is crucial for managing heat in electronic devices, influencing their performance and reliability. Thermal conductivity is a measure of a material's ability to conduct heat, represented by the symbol 'k' and expressed in watts per meter-kelvin ($\text{W}/\text{m}\cdot\text{K}$). In semiconductors, thermal conductivity plays a vital role in determining how effectively heat is dissipated from devices such as transistors, diodes, and integrated circuits. Efficient thermal management is essential to prevent overheating, which can lead to device failure or reduced performance.

Semiconductors generally have moderate thermal conductivity compared to metals and insulators. Here are some typical values for common semiconductor materials at room temperature:

- **Silicon (Si):** Approximately **149 $\text{W}/\text{m}\cdot\text{K}$** .
- **Gallium Arsenide (GaAs):** Ranges from **50 to 100 $\text{W}/\text{m}\cdot\text{K}$** , depending on purity and temperature.
- **Silicon Carbide (SiC):** Can reach up to **490 $\text{W}/\text{m}\cdot\text{K}$** , making it suitable for high-temperature

applications.

- **Boron Arsenide (BAs):** Exhibits exceptionally high thermal conductivity, potentially reaching **1300 W/m·K**, which is comparable to some metals.

5. Dielectric Constant

The dielectric constant of a semiconductor indicates its ability to store electrical energy in an electric field. A moderately high dielectric constant is desirable as it improves electric field control and capacitance in semiconductor devices.

The dielectric constant (also called relative permittivity) tells you how well a material can store electric energy when an electric field is applied.

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}$$

where:

- ϵ = permittivity of the material
- ϵ_0 = permittivity of free space

High dielectric constant (high-k) is good for the gate oxide in devices like (Metal–Oxide–Semiconductor Field-Effect Transistor) MOSFET because it: Gives better control of current, Reduces leakage and Improves device performance.

Low dielectric constant (low-k) is good for wires (interconnects) inside chips because it: Reduces unwanted capacitance, Makes signals travel faster, Saves power

6. Resistivity (ρ)

Resistivity is the opposition to current flow in a semiconductor. Semiconductors require moderate and controllable resistivity, which can be adjusted by doping to allow proper device operation.

It determines current flow, used to classify materials (conductor / semiconductor / insulator) and critical for device design and IC fabrication

7. Breakdown Voltage

The maximum voltage a material can withstand before electrical failure occurs.

Importance: Critical for power electronics; higher breakdown voltage allows devices to handle larger voltages safely.

Example: Wide band gap materials (like SiC, GaN) have much higher breakdown voltages than silicon.

8. Doping Capability

The ability to introduce controlled impurities (dopants) into a semiconductor to modify its conductivity.

Types:

n-type doping: Adds extra electrons (e.g., phosphorus in silicon).

p-type doping: Creates holes by removing electrons (e.g., boron in silicon).

Importance: Enables the creation of diodes, transistors, and integrated circuits.

[Semiconductor Fabrication: Understanding Its Processes and Importance](#)

Device Fabrication:

Integrated Circuits (ICs) are the foundation of modern electronic systems and play a vital role in computers, communication devices, medical instruments, and automation technologies. They are manufactured through semiconductor device fabrication processes, where microscopic components such as transistors and diodes are integrated onto a single silicon chip. Continuous advancements in fabrication technology have enabled miniaturization, improved performance, and reduced power consumption, making ICs essential for modern technological development.